A JOINT-DISTRIBUTION ROUTE TO FAIR REPRESENTATIONS WITH CONTINUOUS SENSITIVE ATTRIBUTES*

YIJIN NI† AND XIAOMING HUO†

Abstract. Fair representation learning with a continuous sensitive attribute is often operationalized through conditional-distribution discrepancies of the form $\mathbb{E}_S[d(P_{Z|S}, P_Z)]$, as in Generalized Demographic Parity (GDP) and the Expectation of Integral Probability Metrics (EIPM). This conditional route is conceptually natural: it asks whether the representation distribution changes as the sensitive value varies. For continuous S, however, its empirical implementation requires smoothing over S, bandwidth choices, leave-one-out corrections, and repeated conditional-distribution estimates. We propose a joint-distribution route instead. Rather than estimating conditional distributions over a continuum of sensitive values, FRHSIC enforces representation-level independence by penalizing the discrepancy between $P_{Z,S}$ and $P_Z \otimes P_S$ using HSIC. Under characteristic kernels, this joint criterion is zero-equivalent to EIPM and targets the same population independence condition. Beyond zero-level equivalence, we show that empirical HSIC controls a projected empirical conditional-gap surrogate for RKHS downstream heads through a spectral bound, with the unprojected gap controlled under a full-rank condition on the sensitive-attribute Gram matrix. Experiments show competitive fairness–accuracy tradeoffs, transferability across downstream heads, and substantially lower per-epoch computational cost than conditional-estimation-based baselines.

Key words. fair representation learning, continuous sensitive attributes, Hilbert–Schmidt independence criterion, kernel methods, demographic parity

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1. Introduction.

For binary sensitive attributes, fair representation learning can be viewed as a finite conditional-distribution matching problem: one compares $P_{Z|S=0}$ and $P_{Z|S=1}$. For continuous sensitive attributes, this viewpoint leads naturally to integral conditional criteria such as GDP and EIPM, which average discrepancies between $P_{Z|S=s}$ and P_Z. These criteria are conceptually natural, but they operationalize fairness through a difficult object: the conditional family $\{P_{Z|S=s}\}_{s\in\mathcal{S}}$. Estimating this family requires smoothing over S, bandwidth choices, and weighted-empirical corrections.

This paper proposes a different estimation route. Representation-level fairness is the independence condition $Z \perp S$, equivalently $P_{Z,S} = P_Z \otimes P_S$. Therefore, instead of estimating conditional distributions over a continuum of sensitive values, we estimate a joint-vs-product discrepancy directly from paired samples. We instantiate this joint route with the Hilbert–Schmidt Independence Criterion (HSIC), a kernel discrepancy between the joint distribution and the product of marginals.

Background on fair representation learning.. As machine learning becomes increasingly integrated into social decision-making (e.g., hiring, lending, criminal justice), ensuring algorithmic fairness has emerged as an important challenge (Calders and Verwer, 2010). Consider an input random variable X (e.g., applicant data), a protected attribute S (e.g., gender, race, age), and a target variable Y (e.g., loan default). A naïve approach might exclude S from model training; however, when X and S are statistically dependent, models trained on X alone can still perpetuate discrimination

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†H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology, Atlanta, GA 30332-0205 USA (yni64@gatech.edu, huo@gatech.edu).

by leveraging proxy features correlated with S. Fair Representation Learning (FRL) addresses this by learning a representation $Z = h(X)$ that is informative about Y while being independent of S (Zemel et al., 2013; Louizos et al., 2015; Madras et al., 2018); when Z satisfies $Z \perp S$, every downstream predictor $f(Z)$ inherits the fairness property.

1.1. The Integral Framework for Continuous Sensitive Attributes. When S is binary, Demographic Parity (Calders and Verwer, 2010) requires $\widehat{Y} \perp S$, and the induced condition $P_{Z|S=0} \approx P_{Z|S=1}$ is a two-sample testing problem with well-studied estimators based on Integral Probability Metrics (IPMs).

When S is continuous (e.g., age, income), the two-sample IPM does not directly apply: the conditional distributions $\{P_{Z|S=s}\}_{s\in\mathcal{S}}$ form an uncountable family with at most one observation per value of s. Existing approaches replace it with the *integral framework*, an average of a divergence between the conditional and marginal distributions of Z:

$$\mathcal{I}_d(Z; S) := \mathbb{E}_S\big[d(P_{Z|S}, P_Z)\big]\,, \tag{1.1}$$

for a chosen discrepancy d. This is conceptually natural: it directly measures how the distribution of Z shifts as s varies. Several widely-used continuous-attribute fairness criteria are instances:

- **Generalized Demographic Parity (GDP)** (Jiang et al., 2022): d is a first-moment difference; estimated via Nadaraya–Watson kernel smoothing on S.
- **Expectation of IPMs (EIPM/FREM)** (Kong et al., 2025): d is an IPM (a kernel MMD instance); estimated via a weighted-empirical procedure with smoothing bandwidth γ_n and leave-one-out corrections.
- **Mutual information** (Cho et al., 2020): d is a KL divergence; estimated via density or copula approximations.

A related criterion, **kernel HGR / NLD** (Mary et al., 2019; Grari et al., 2022), replaces the integral (1.1) by a global supremum, $\rho^*_{\mathrm{HGR}} := \sup_{f,g} \mathrm{Corr}(f(Z), g(S))$, and its empirical implementation reduces to regularized kernel CCA on (Z, S). It does not require conditional smoothing on S, but its estimation is its own challenge: empirical kCCA / kHGR is unstable without the regularization, and selecting the regularization parameters introduces a separate model-selection step.

The $\mathcal{I}_d$ instances above (GDP, EIPM, MI) share, operationally, conditional-distribution machinery on S: a smoothing bandwidth γ_n, leave-one-out or weighted-empirical corrections, and a nonparametric convergence rate that is slower than $n^{-1/2}$ (e.g., $O_p(n^{-2/5})$ for EIPM under their optimal bandwidth choice (Kong et al., 2025)). These are practical costs of estimating an integral over conditional distributions; they do not reflect any conceptual problem with the integral form itself. Empirical kHGR has different practical costs (regularized operator estimation), unrelated to conditional smoothing.

1.2. A Joint-Distribution Route via HSIC. Independence $Z \perp S$ is, equivalently, a property of the *joint* distribution: $P_{Z,S} = P_Z \otimes P_S$. A discrepancy between the joint and the product of marginals therefore offers a natural alternative route to the same population target as (1.1), one that does not require constructing the conditional family.

The Hilbert–Schmidt Independence Criterion (HSIC) of Gretton et al. (2005) is one such discrepancy. Under characteristic kernels, $\mathrm{HSIC}(Z, S) = 0$ if and only if

$Z \perp S$, and the criterion is computable as a closed-form V-statistic

$$\widehat{\mathrm{HSIC}}(Z, S) = \frac{1}{n^2}\,\mathrm{tr}(KHLH),$$

in $O(n^2)$ time with no smoothing bandwidth on S and no leave-one-out correction. As an estimator of its target functional HSIC, it converges at the standard parametric rate $O_p(n^{-1/2})$ (Gretton et al., 2005); the smoothed estimator of EIPM converges at $O_p(n^{-2/5})$ (Kong et al., 2025). These rates compare different empirical objectives, not the same target — they are not a claim that HSIC estimates EIPM more accurately, nor that EIPM is conceptually incorrect.

Two consequences of this perspective bear on FRL objective design:

1. **Same population target.** Under characteristic kernels, HSIC vanishes on the same joints $P_{Z,S} = P_Z \otimes P_S$ as EIPM and MI (so for these two, HSIC is genuinely zero-equivalent), while for GDP we only claim the one-way implication $\mathrm{HSIC}(Z, S) = 0 \Rightarrow Z \perp S \Rightarrow \Delta_{\mathrm{GDP}_\phi}(g) = 0$ for every prediction head $\phi \circ g$ (Proposition 3.6; full proofs in Appendix SM1 of the supplement). The converse fails in general: a single fixed head being fair is consistent with nontrivial Z–S dependence on other heads. Away from zero the penalties are different functionals of $P_{Z,S}$, and we do not claim quantitative dominance.
2. **No conditional-density estimation.** Computing $\widehat{\mathrm{HSIC}}$ requires only the joint sample $\{(Z_i, S_i)\}$ and two kernel matrices; no smoothing on S and no leave-one-out is needed. This is the source of HSIC's $O(n^2)$ closed form and of FRHSIC's per-epoch speedup over an EIPM-based objective.

The integral form (1.1) is operationally demanding for continuous S because the conditional family must be estimated nonparametrically; the joint-distribution route bypasses that estimation step while targeting the same population independence condition.

Why the joint route matters.. The distinction between the conditional route and the joint route is not merely notational. Conditional criteria such as GDP and EIPM begin from the family $\{P_{Z|S=s}\}_s$ and therefore require smoothing or weighting over the sensitive coordinate when S is continuous. The joint route begins from the equivalent independence condition $P_{Z,S} = P_Z \otimes P_S$, which can be estimated directly from paired samples. HSIC is the kernel statistic we use to instantiate this route. Its role in this paper is to turn the joint formulation into an efficient training objective and to connect that objective back to the conditional deviations measured by GDP/EIPM.

1.3. Summary of Contributions.

1. **A joint-distribution formulation for continuous-sensitive-attribute FRL.** We propose to formulate representation-level fairness with continuous sensitive attributes through the joint condition $P_{Z,S} = P_Z \otimes P_S$, rather than through explicit estimation of the conditional family $\{P_{Z|S=s}\}_s$.
2. **Theoretical bridges back to conditional fairness.** We show that the joint route preserves the population fairness target: HSIC is zero-equivalent to EIPM under characteristic kernels (Proposition 3.9 and Theorem 3.1), and HSIC $= 0$ implies $\Delta_{\mathrm{GDP}_\phi}(g) = 0$ for every downstream head (Proposition 3.6). Beyond zero-level equivalence, empirical HSIC controls a projected empirical conditional-gap surrogate through a spectral inequality (Theorem 3.4; unprojected bound under a full-rank condition in Corollary 3.5).
3. **Algorithmic consequence.** We instantiate the joint route as FRHSIC, a mini-batch fair-representation objective that avoids conditional smoothing on

S, leave-one-out corrections, and repeated weighted-MMD computations.

4. **Empirical consequence.** On synthetic and real datasets, FRHSIC attains competitive fairness–accuracy tradeoffs, transfers across downstream heads, and trains substantially faster per epoch than conditional-estimation-based baselines (Section 4).

TABLE 1

Two estimation routes to representation-level fairness with continuous sensitive attributes. The conditional route estimates discrepancies involving $P_{Z|S=s}$; the joint route estimates a discrepancy between $P_{Z,S}$ and $P_Z \otimes P_S$ directly from paired samples.

Route	Object	Burden	Methods
Conditional-distribution route	$\mathbb{E}_S[d(P_{Z\|S}, P_Z)]$	Estimate/smooth $P_{Z\|S=s}$ over S	GDP, EIPM/FREM
Joint-distribution route	$d(P_{Z,S}, P_Z \otimes P_S)$	Estimate joint-vs-product discrepancy from paired samples	FRHSIC

2. Preliminaries.

2.1. Notation. Let $X \in \mathcal{X}$ denote the input random vector, $S \in \mathcal{S} \subseteq \mathbb{R}$ the (continuous) sensitive attribute, and $Y \in \mathcal{Y}$ the target variable. A representation function $h : \mathcal{X} \to \mathcal{Z}$ maps the input to a representation space, yielding $Z = h(X)$. A prediction head $f : \mathcal{Z} \to \mathcal{Y}$ maps the representation to predictions, so that $\widehat{Y} = f(Z) = f \circ h(X)$.

We denote by P_Z the marginal distribution of Z, by $P_{Z|S=s}$ the conditional distribution of Z given $S = s$, and by $P_{Z,S}$ the joint distribution. For a reproducing kernel Hilbert space (RKHS) $\mathcal{F}$ with kernel k, the mean embedding of a distribution P is $\mu_P = \mathbb{E}_{X\sim P}[k(X, \cdot)] \in \mathcal{F}$.

2.2. Fair Prediction Models.
Demographic Parity (DP). For binary $S \in \{0, 1\}$, the predictor $\widehat{Y}$ satisfies DP if $\widehat{Y} \perp S$, which for binary $\widehat{Y}$ reduces to $\mathbb{E}[\widehat{Y} \mid S = 0] = \mathbb{E}[\widehat{Y} \mid S = 1]$.

Generalized Demographic Parity (GDP). For continuous S, Jiang et al. (2022) define GDP as

$$\Delta_{\mathrm{GDP}_\phi}(g) = \mathbb{E}_S \left|\mathbb{E}_X[\phi \circ g(X) \mid S] - \mathbb{E}_X[\phi \circ g(X)]\right|, \tag{2.1}$$

where $\circ$ denotes function composition, i.e., $(\phi \circ g)(x) = \phi(g(x))$; the right-hand side is estimated via kernel smoothing (Nadaraya, 1964; Watson, 1964), and reduces to DP when S is binary.

Expectation of IPM (EIPM). Kong et al. (2025) define the EIPM as

$$\mathrm{EIPM}_{\mathcal{V}}(Z; S) := \mathbb{E}_S[\mathrm{IPM}_{\mathcal{V}}(P_{Z|S}, P_Z)],$$

where $\mathrm{IPM}_{\mathcal{V}}(P, Q) = \sup_{f\in\mathcal{V}} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]|$. They show that controlling EIPM controls GDP.

2.3. Fair Representation Learning. The goal of FRL is to learn a representation $Z = h(X)$ such that Z is independent of S (fairness) while being informative about Y (sufficiency). For binary S, this reduces to matching the conditional distributions: $P_{Z|S=0} \approx P_{Z|S=1}$. For continuous S, the requirement generalizes to

$$P_{Z|S=s} \approx P_Z, \quad \forall s \in \mathcal{S}, \tag{2.2}$$

which is equivalent to requiring $Z \perp S$.

2.4. Integral Probability Metrics and MMD. The Maximum Mean Discrepancy (MMD) between two distributions P and Q is the IPM over the unit ball of an RKHS $\mathcal{F}$:

$$\mathrm{MMD}(P, Q) = \sup_{\|f\|_{\mathcal{F}} \leq 1} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]| = \|\mu_P - \mu_Q\|_{\mathcal{F}}.$$

When k is a characteristic kernel, $\mathrm{MMD}(P, Q) = 0$ if and only if $P = Q$ (Gretton et al., 2012).

2.5. Hirschfeld–Gebelein–Rényi Maximal Correlation. The Hirschfeld–Gebelein–Rényi (HGR) maximal correlation between random variables Z and S is

$$\mathrm{HGR}(Z, S) := \sup_{f,g} \mathrm{Corr}(f(Z), g(S)),$$

where the supremum is over measurable functions f, g with $\mathbb{E}[f(Z)] = \mathbb{E}[g(S)] = 0$ and unit variance. HGR characterizes independence: $\mathrm{HGR}(Z, S) = 0 \iff Z \perp S$ (Rényi, 1959). The kernelized variant kHGR, used in fair representation learning by Mary et al. (2019) and Grari et al. (2022), restricts f, g to RKHS unit balls; under characteristic kernels, $\mathrm{kHGR}(Z, S) = 0 \iff Z \perp S$ as well.

2.6. Mutual Information. The mutual information between Z and S is

$$I(Z; S) = \mathbb{E}_S\big[\mathrm{KL}(P_{Z|S} \| P_Z)\big] = \int \log \frac{p_{Z,S}}{p_Z\, p_S}\, dP_{Z,S},$$

where the second form assumes joint and marginal densities exist. Mutual information characterizes independence: $I(Z; S) = 0 \iff Z \perp S$ (Cover and Thomas, 2006, Theorem 2.6.3).

2.7. The Integral Framework. We use the term *integral framework* for any fairness functional of the form

$$\mathcal{I}_d(Z; S) := \mathbb{E}_S\big[d(P_{Z|S}, P_Z)\big]$$

for some discrepancy d between distributions. This formulation is well-defined for Polish $\mathcal{Z}, \mathcal{S}$ once a regular conditional probability $\{P_{Z|S=s}\}_{s \in \mathcal{S}}$ is selected; existence is guaranteed by the disintegration theorem (Kallenberg, 2002, Theorem 5.4). GDP, EIPM, and mutual information instantiate $\mathcal{I}_d$ for d equal to a first-moment difference, an IPM, and the KL divergence, respectively. kHGR/NLD is a global-dependence criterion estimated through regularized kernel CCA. It avoids conditional smoothing on S but introduces separate regularization and finite-sample stability issues. We include kHGR (via the ADV implementation of Grari et al. 2022) as a global-dependence baseline; our main conditional-estimation comparison concerns GDP and EIPM.

2.8. Structural Causal Models and Counterfactual Fairness. Following Pearl (2009) and Kusner et al. (2017), a Structural Causal Model (SCM) over (X, S, Y) specifies structural equations

$$S = U_S, \qquad X = f_X(S, U_X), \qquad Y = f_Y(X, S, U_Y),$$

with jointly independent exogenous noise U_X, U_S, U_Y. The counterfactual prediction $\widehat{Y}_{S\leftarrow s'}$ is the value of $\widehat{Y}$ obtained by replacing $S = U_S$ with $S = s'$ in the structural equations and propagating downstream. A predictor $\widehat{Y}$ is *counterfactually fair* (Kusner et al., 2017) if, for every observable $(X = x, S = s)$ and every alternative s',

$$\widehat{Y}_{S\leftarrow s'} = \widehat{Y}_{S\leftarrow s} \quad \text{a.s. given } X = x, S = s.$$

3. FRHSIC: Fair Representation Learning via a Joint-Distribution Route. This section develops the joint-distribution route in four steps: (i) we first recall the conditional-estimation bottleneck that motivates the alternative; (ii) we introduce HSIC as the joint-independence criterion that bypasses it; (iii) we establish a zero-equivalence bridge to EIPM and a population implication for GDP; and (iv) we prove a finite-sample conditional-gap control statement that shows the joint criterion still governs the conditional deviations measured by integral fairness metrics. The FRHSIC objective, validation-based λ selection, and Equal Opportunity extension are then assembled from these ingredients.

3.1. The conditional-estimation bottleneck. Recall the integral framework $\mathcal{I}_d(Z; S) = \mathbb{E}_S[d(P_{Z|S}, P_Z)]$ from Section 1.1. For continuous S, every estimator of $\mathcal{I}_d$ must access the conditional family $\{P_{Z|S=s}\}_{s\in\mathcal{S}}$, an uncountable object with at most one observation per s. Standard implementations therefore choose a smoothing bandwidth on S (e.g., Nadaraya–Watson for GDP, weighted-empirical kernel smoothing for EIPM/FREM), use leave-one-out corrections for tractability, and inherit a nonparametric estimation rate slower than $n^{-1/2}$. These operational costs are not pathologies of the integral form itself; they reflect the conditional-distribution machinery the form invokes. The remainder of this section develops a joint-distribution alternative that targets the same population independence condition without this machinery, and recovers conditional-gap control through theoretical bridges rather than nonparametric smoothing.

3.2. The joint-independence alternative. Let $k_{\mathcal{Z}} : \mathcal{Z} \times \mathcal{Z} \to \mathbb{R}$ and $k_{\mathcal{S}} : \mathcal{S} \times \mathcal{S} \to \mathbb{R}$ be characteristic kernels with corresponding RKHSs $\mathcal{F}_{\mathcal{Z}}$ and $\mathcal{F}_{\mathcal{S}}$, respectively.

The Hilbert–Schmidt Independence Criterion (Gretton et al., 2005) measures the dependence between Z and S as

$$\text{HSIC}(Z, S) = \|\mu_{Z,S} - \mu_Z \otimes \mu_S\|^2_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}}, \tag{3.1}$$

where $\mu_{Z,S}$ is the joint mean embedding and $\mu_Z \otimes \mu_S$ is the product of the marginal embeddings. With characteristic kernels, $\text{HSIC}(Z, S) = 0$ if and only if $Z \perp S$.

We define the conditional MMD divergence between $P_{Z|S=s}$ and P_Z as

$$\gamma_{k_{\mathcal{Z}}}(Z|S = s, Z) = \text{MMD}(P_{Z|S=s}, P_Z) = \left\|\mu_{Z|S=s} - \mu_Z\right\|_{\mathcal{F}_{\mathcal{Z}}}, \tag{3.2}$$

where $\mu_{Z|S=s} = \mathbb{E}[k_{\mathcal{Z}}(Z, \cdot)|S = s]$ is the conditional mean embedding.

3.3. Bridge I: zero-equivalence with EIPM and implication for GDP. We now present the first theoretical bridge connecting the joint route back to conditional fairness criteria. HSIC and EIPM are zero-equivalent under characteristic kernels; mutual information is zero exactly at independence; and HSIC = 0 implies $\Delta_{\mathrm{GDP}_\phi}(g) = 0$ for every downstream head. A single fixed-head GDP constraint is weaker and does not characterize independence. For HSIC, we assume throughout that the product kernel $k_{\mathcal{Z}} \otimes k_{\mathcal{S}}$ is characteristic to the relevant joint and product measures on $\mathcal{Z} \times \mathcal{S}$; this holds under standard conditions for the Gaussian kernels used in our experiments.

THEOREM 3.1 (HSIC and Expected MMD). *Suppose $k_{\mathcal{Z}}$ and $k_{\mathcal{S}}$ are bounded, i.e., $k_{\mathcal{Z}}(z,z) \le \kappa_{\mathcal{Z}}$ and $k_{\mathcal{S}}(s,s) \le \kappa_{\mathcal{S}}$ for all $z \in \mathcal{Z}$, $s \in \mathcal{S}$. Then HSIC admits the decomposition*

$$\mathrm{HSIC}(Z,S) = \mathbb{E}_{S,S'}\left[\left\langle \mu_{Z|S} - \mu_Z, \mu_{Z|S'} - \mu_Z \right\rangle_{\mathcal{F}_{\mathcal{Z}}} \cdot k_{\mathcal{S}}(S,S')\right], \tag{3.3}$$

where S, S' are independent copies. Moreover, HSIC is bounded by the squared expected MMD:

$$\mathrm{HSIC}(Z,S) \le \kappa_{\mathcal{S}} \cdot \left(\mathbb{E}_S[\gamma_{k_{\mathcal{Z}}}(Z \mid S, Z)]\right)^2. \tag{3.4}$$

Proof. See Appendix SM1.1 of the supplement. ∎

Remark 3.2. Equation (3.3) reveals that HSIC aggregates the pairwise similarity of conditional deviations $\mu_{Z|S} - \mu_Z$ weighted by the kernel similarity $k_{\mathcal{S}}(S,S')$ between the corresponding sensitive attribute values. The bound in Equation (3.4) shows that HSIC is upper-bounded by the squared expected MMD (i.e., the squared EIPM when $\mathcal{V}$ is the unit ball of $\mathcal{F}_{\mathcal{Z}}$). Note that this bound goes from EIPM to HSIC (small EIPM implies small HSIC), not the reverse; the useful reverse direction—HSIC controls a finite-sample empirical conditional-gap quantity that dominates the squared empirical absolute prediction gap for any RKHS prediction head—is established separately by Theorem 3.4 (with the population-level link in Proposition 3.6). The inequality compares the scale of the two population penalties; it does not imply monotone dominance away from zero. HSIC and EIPM are zero-equivalent under characteristic kernels, but away from zero they weight conditional deviations differently: EIPM averages conditional MMD magnitudes, while HSIC measures a covariance-weighted joint-vs-product discrepancy.

For the GDP connection, denote the composite function $f = \phi \circ g$ and define the population conditional difference

$$\mathcal{D}_f(S) := \mathbb{E}_Z[f(Z) \mid S] - \mathbb{E}_Z[f(Z)].$$

The next theorem controls the corresponding empirical centered prediction $\hat{\delta}_{f,i} := f(Z_i) - n^{-1}\sum_{j=1}^{n} f(Z_j)$ at the observed sensitive values.

PROPOSITION 3.3 (HSIC controls kernel-smoothed conditional deviations). *Suppose $k_{\mathcal{Z}}$ and $k_{\mathcal{S}}$ are bounded measurable kernels with $k_{\mathcal{Z}}(z,z) \le \kappa_{\mathcal{Z}}$ and $k_{\mathcal{S}}(s,s) \le \kappa_{\mathcal{S}}$ for all $z \in \mathcal{Z}$, $s \in \mathcal{S}$, so that the population mean embeddings $\mu_Z = \mathbb{E}_Z[k_{\mathcal{Z}}(Z,\cdot)] \in \mathcal{F}_{\mathcal{Z}}$ and $\mu_S = \mathbb{E}_S[k_{\mathcal{S}}(S,\cdot)] \in \mathcal{F}_{\mathcal{S}}$ exist as Bochner integrals. Define the centered cross-covariance operator $C_{SZ} : \mathcal{F}_{\mathcal{Z}} \to \mathcal{F}_{\mathcal{S}}$ via*

$$C_{SZ} := \mathbb{E}\big[(k_{\mathcal{S}}(S,\cdot) - \mu_S) \otimes (k_{\mathcal{Z}}(Z,\cdot) - \mu_Z)\big],$$

identified with a Hilbert–Schmidt operator through the tensor-product representation. Then $\mathrm{HSIC}(Z,S)=\|C_{SZ}\|_{\mathrm{HS}}^2$ *(Gretton et al., 2005), and for any* $f\in\mathcal{F}_{\mathcal{Z}}$*,*

$$\|C_{SZ}f\|_{\mathcal{F}_{\mathcal{S}}}^2 \;\le\; \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2\,\mathrm{HSIC}(Z,S). \tag{3.5}$$

Equivalently, writing $m_f(s):=\mathbb{E}[f(Z)\mid S=s]-\mathbb{E}[f(Z)]$ *for the centered conditional expectation,*

$$\big\|\mathbb{E}_S[m_f(S)\,k_{\mathcal{S}}(S,\cdot)]\big\|_{\mathcal{F}_{\mathcal{S}}}^2 \;\le\; \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2\,\mathrm{HSIC}(Z,S).$$

Proof. See Appendix SM1.2 of the supplement. ∎

Proposition 3.3 controls the population kernel-smoothed conditional deviation by HSIC, complementing the finite-sample spectral statement in Theorem 3.4 below. The first targets the smoothed object $\mathbb{E}_S[m_f(S)\,k_{\mathcal{S}}(S,\cdot)]$; the second targets the empirical pointwise gap vector $(\hat\delta_{f,i})_{i=1}^n$. Vanishing HSIC makes both $\|C_{SZ}f\|_{\mathcal{F}_{\mathcal{S}}}^2$ (Proposition 3.3) and the projected empirical conditional-gap surrogate $\frac{1}{n}\sum_i(P_{\widetilde L}\boldsymbol{\delta}_f)_i^2$ (Theorem 3.4) zero for every $f\in\mathcal{F}_{\mathcal{Z}}$. The converse direction—recovering independence from vanishing conditional-deviation quantities—requires considering a sufficiently rich class of test functions or kernels; a single fixed downstream head is not enough. Neither bound implies a uniform pointwise control on $m_f(s)$ at every s without further smoothing assumptions.

3.4. Bridge II: conditional-gap control. The next theorem strengthens the population bridge to a finite-sample statement: empirical HSIC controls a projected empirical conditional-gap surrogate, and under a full-rank condition on $\widetilde L$, the projection is the identity.

THEOREM 3.4 (Empirical conditional-gap control by HSIC, projected version). *Let* $\{(Z_i,S_i)\}_{i=1}^n$ *be a training or held-out sample, let* $K,L\in\mathbb{R}^{n\times n}$ *be the Gram matrices on* Z *and* S *with* $K_{ij}=k_{\mathcal{Z}}(Z_i,Z_j)$ *and* $L_{ij}=k_{\mathcal{S}}(S_i,S_j)$*, and let* $\widetilde K=HKH$*,* $\widetilde L=HLH$*, where* $H=I_n-n^{-1}\mathbf{1}\mathbf{1}^\top$*. Let* $r:=\mathrm{rank}(\widetilde L)$ *(so* $1\le r\le n-1$*), and let* $P_{\widetilde L}$ *denote the orthogonal projection onto the column span of* $\widetilde L$*. Define*

$$\hat\lambda_S:=\lambda_{\min}^{+}(n^{-1}\widetilde L),$$

the smallest positive eigenvalue of $n^{-1}\widetilde L$*. For* $f\in\mathcal{F}_{\mathcal{Z}}$*, define* $\bar f_n:=n^{-1}\sum_{j=1}^n f(Z_j)$*,* $\hat\delta_{f,i}:=f(Z_i)-\bar f_n$*, and let* $(P_{\widetilde L}\boldsymbol{\delta}_f)_i$ *denote the* i*-th coordinate of the projection of* $\boldsymbol{\delta}_f:=(\hat\delta_{f,1},\dots,\hat\delta_{f,n})^\top$ *onto the column span of* $\widetilde L$*. Then*

$$\frac{1}{n}\sum_{i=1}^n\big(P_{\widetilde L}\,\boldsymbol{\delta}_f\big)_i^2 \;\le\; \frac{\|f\|_{\mathcal{F}_{\mathcal{Z}}}^2}{\hat\lambda_S}\,\widehat{\mathrm{HSIC}}(Z,S). \tag{3.6}$$

Proof. See Appendix SM1.3 of the supplement. ∎

The projection isolates the part of the prediction-deviation vector that lies in the directions on which $\widetilde L$ has positive eigenvalues; the kernel of $\widetilde L$ contains the constant direction $\mathbf{1}$ but may also contain additional directions arising from repeated or ordinal S values.

COROLLARY 3.5 (Unprojected bound under full rank). *If* $\widetilde L$ *has rank exactly* $n-1$ *(e.g., for strictly positive definite kernels such as the Gaussian kernel on distinct sample points; note that being characteristic alone does not in general guarantee* *finite-sample strict positive definiteness), then $P_{\widetilde{L}}$ projects onto the orthogonal complement of* $\operatorname{span}\{\mathbf{1}\}$*, and the unprojected centered-deviation vector $\boldsymbol{\delta}_f$ already lies in this complement (since $\sum_i \hat{\delta}_{f,i} = 0$). Hence* (3.6) *reduces to*

$$\frac{1}{n}\sum_{i=1}^{n}\hat{\delta}_{f,i}^2 \;\le\; \frac{\|f\|_{\mathcal{F}_{\mathcal{Z}}}^2}{\hat{\lambda}_S}\,\widehat{\mathrm{HSIC}}(Z,S),$$

and consequently, for the empirical absolute prediction gap $\hat{\Delta}_f := n^{-1}\sum_{i=1}^{n}|\hat{\delta}_{f,i}|$,

$$\hat{\Delta}_f^2 \;\le\; \frac{1}{n}\sum_{i=1}^{n}\hat{\delta}_{f,i}^2 \;\le\; \frac{\|f\|_{\mathcal{F}_{\mathcal{Z}}}^2}{\hat{\lambda}_S}\,\widehat{\mathrm{HSIC}}(Z,S).$$

Interpretation.. Theorem 3.4 explains why the joint route is relevant to conditional fairness metrics. Although FRHSIC does not explicitly estimate $P_{Z|S=s}$, small empirical HSIC forces the projected empirical prediction-gap vector to be small. Under the full-rank sensitive-kernel condition, this gives direct control of the empirical squared conditional-gap surrogate. The spectral factor quantifies the difficulty of converting joint-dependence control into pointwise conditional-gap control over the observed sensitive values. Thus, the joint route bypasses conditional estimation without abandoning the conditional deviations that GDP/EIPM are designed to measure.

PROPOSITION 3.6 (Population HSIC and GDP). *Under characteristic kernels $k_{\mathcal{Z}}, k_{\mathcal{S}}$,* $\mathrm{HSIC}(Z,S) = 0$ *if and only if $Z \perp S$ (Proposition 3.9). When $Z \perp S$, every prediction head $f \circ g \circ h$ satisfies $\Delta_{\mathrm{GDP}_\phi}(g) = 0$, since $\mathbb{E}[\phi \circ g(X) \mid S] = \mathbb{E}[\phi \circ g(X)]$ a.s. Hence* $\mathrm{HSIC}(Z,S) = 0 \implies \Delta_{\mathrm{GDP}_\phi}(g) = 0$ *at the population level.*

Remark 3.7. This is a finite-sample conditional-gap control statement. The quantities $\hat{\delta}_{f,i}$ are centered empirical prediction deviations at the observed sensitive values; they are not a fully smoothed estimator of the population conditional expectation $\mathbb{E}[f(Z) \mid S = s]$ unless an additional smoothing scheme on $\mathcal{S}$ is introduced. The spectral factor $\hat{\lambda}_S^{-1}$ quantifies the difficulty of converting joint-dependence control (small $\widehat{\mathrm{HSIC}}$) into pointwise conditional-gap control over the observed sample. The qualitative limit $\widehat{\mathrm{HSIC}} \to 0 \Rightarrow (P_{\widetilde{L}}\boldsymbol{\delta}_f)_i \equiv 0$ for any f holds without any spectral-gap assumption: the V-statistic $\widehat{\mathrm{HSIC}} = n^{-2}\mathrm{tr}(\widetilde{K}\widetilde{L})$ vanishes precisely when the centered Gram matrices $\widetilde{K}, \widetilde{L}$ are mutually orthogonal in trace, which forces the prediction-deviation vector to lie in the kernel of $\widetilde{L}$ and hence have zero projection onto the column span.

Remark 3.8 (Bandwidth scaling of $\hat{\lambda}_S$). With the normalization $\hat{\lambda}_S = \lambda_{\min}^{+}(n^{-1}\widetilde{L})$, the spectral constant reflects the conditioning of the centered Gram matrix on S. As the sensitive-kernel bandwidth tends to zero and $L \to I_n$, $\widetilde{L} \to H$ (whose positive eigenvalues are all 1), and $\hat{\lambda}_S \to 1/n$. As the bandwidth tends to infinity and $L \to \mathbf{1}\mathbf{1}^\top$, $\widetilde{L} \to 0$ and $\hat{\lambda}_S \to 0$. Very large bandwidths therefore make the conditional-gap bound loose, while extremely small bandwidths give a finite-sample point-mass notion of conditioning; the median heuristic selects an intermediate regime. Standard results on Gaussian-kernel Gram matrices for samples from densities with compact support (e.g., Koltchinskii and Giné, 2000) confirm a polynomial dependence on σ between these extremes; in our experiments at $n = 1500$ this yields $\hat{\lambda}_S \sim 10^{-9}$–$10^{-7}$ across the bandwidth sweep (Section SM7.2).

Relation to kernel HGR. The Hirschfeld–Gebelein–Rényi (HGR) maximal correlation (Rényi, 1959) characterizes independence under broad richness assumptions on the function classes; its kernelized variant kHGR (Mary et al., 2019) replaces the classes with RKHS unit balls. Population HGR/kHGR and HSIC are therefore both zero precisely when $Z \perp S$ under suitable kernel choices. Empirical kHGR estimation, however, typically requires regularized kernel CCA (see, e.g., Fukumizu et al. 2007); we do not rely on kHGR for the FRHSIC guarantee or appeal to an empirical kHGR sandwich bound. A regularized kHGR statement is recorded in Appendix SM3 for completeness.

We next establish that HSIC and EIPM detect the same set of fair representations at the population level, so the substitution loses no expressive power—and, as a reference point, the empirical HSIC estimator runs in $O(n^2)$ time versus $O(n^3)$ for the EIPM estimator (Section 3), yielding the computational advantage without weakening the fairness criterion.

PROPOSITION 3.9 (Population-level equivalence of HSIC and EIPM). *Let $k_{\mathcal{Z}}$ and $k_{\mathcal{S}}$ be characteristic kernels with corresponding RKHSs $\mathcal{F}_{\mathcal{Z}}$ and $\mathcal{F}_{\mathcal{S}}$, and let $\mathcal{V} = \{f \in \mathcal{F}_{\mathcal{Z}} : \|f\|_{\mathcal{F}_{\mathcal{Z}}} \leq 1\}$ be the unit ball in $\mathcal{F}_{\mathcal{Z}}$. Then*

$$\mathrm{HSIC}(Z, S) = 0 \quad \Longleftrightarrow \quad \mathrm{EIPM}_{\mathcal{V}}(Z; S) = 0.$$

Proof. See Appendix SM1.4 of the supplement. ∎

Remark 3.10. Proposition 3.9 confirms that HSIC and EIPM are not strictly comparable in strength—they are two characterizations of the same independence condition $Z \perp S$. The substitution we propose therefore changes only the estimation procedure, not the fairness criterion itself.

We next formalize the conceptual claim of Section 1 that HSIC takes a joint-distribution route to the same population independence target, while the integral framework constructs the same target through the conditional family $\{P_{Z|S=s}\}$. HSIC and EIPM (and the analogous integral measures $\mathcal{I}_d$) are zero-equivalent at the population level but are different nonzero functionals; consequently, the rate comparison below is between estimators of *different* objectives that happen to share their zero set. Two observations substantiate this structural distinction: (i) $\widehat{\mathrm{HSIC}}$ is a plug-in joint V-statistic on $P_{Z,S}$, while standard estimators of $\mathcal{I}_d$ smooth on the conditional family $\{P_{Z|S=s}\}$ (Remark 3.11); (ii) the parametric estimation rate of $\widehat{\mathrm{HSIC}}$ versus the nonparametric-rate estimators of $\mathcal{I}_d$ is the empirical reflection of (i) at the level of estimators of these distinct functionals; and (iii) zero HSIC delivers the same per-subgroup fairness guarantee that integral measures aim to enforce.

Remark 3.11 (Plug-in joint V-statistic vs. conditional smoothing). $\mathrm{HSIC}(Z, S)$ admits a closed-form expression in terms of the joint distribution $P_{Z,S}$ and its marginal projections only, namely $\mathrm{HSIC}(Z, S) = \|\mu_{Z,S} - \mu_Z \otimes \mu_S\|^2_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}}$. Its empirical estimator $\widehat{\mathrm{HSIC}} = n^{-2}\,\mathrm{tr}(KHLH)$ is therefore a plug-in joint V-statistic that uses only the empirical joint sample. The standard expression of $\mathcal{I}_d(Z; S) = \mathbb{E}_S[d(P_{Z|S}, P_Z)]$, by contrast, evaluates a discrepancy on the conditional family $\{P_{Z|S=s}\}_{s\in\mathcal{S}}$, and standard estimators construct $\hat{P}_{Z|S=s}$ via kernel smoothing on S. Both quantities are Borel functionals of $P_{Z,S}$ once a regular conditional version is fixed; the practical difference is in their estimators (joint V-statistic vs. conditional-smoothing-based estimator).

Remark 3.12 (Estimation-rate comparison for the two objectives). The reliance of standard $\mathcal{I}_d$ estimators on the conditional family is reflected in their rates: every such estimator involves nonparametric estimation of $P_{Z|S=\cdot}$ and inherits the corresponding minimax rate, which for β-Hölder conditional densities on $\mathbb{R}$ is $O(n^{-2\beta/(2\beta+1)})$ (Tsybakov, 2009, Chapter 2)—a nonparametric rate, slower than parametric. By contrast, $\widehat{\mathrm{HSIC}}_n = n^{-2}\,\mathrm{tr}(KHLH)$ is a bounded V-statistic on the joint sample and converges at the parametric rate $O_p(n^{-1/2})$ for any bounded characteristic kernel (Proposition 3.15). The rate comparison is therefore between estimators of two different (but population-zero-equivalent) objectives: the joint-functional HSIC admits a parametric-rate plug-in estimator, while standard estimators of integral measures $\mathcal{I}_d$ inherit the nonparametric rate of conditional-density estimation. We view this as a quantitative companion to the estimator-level asymmetry of Remark 3.11: the joint route and the conditional route target the same population independence condition but as different nonzero functionals, and their plug-in estimators inherit different convergence rates.

THEOREM 3.13 (Subgroup-fairness equivalence). *Let $\mathcal{Z}, \mathcal{S}$ be Polish spaces, let $k_\mathcal{Z}, k_\mathcal{S}$ be characteristic kernels, and let (Z, S) be a pair of random variables on $\mathcal{Z} \times \mathcal{S}$ with bounded $k_\mathcal{Z}, k_\mathcal{S}$. The following four conditions are equivalent:*

(i) $\mathrm{HSIC}(Z, S) = 0$.

(ii) $Z \perp S$, *i.e.,* $P_{Z,S} = P_Z \otimes P_S$.

(iii) For some (equivalently, every) regular conditional probability $\{P_{Z|S=s}\}_{s\in\mathcal{S}}$, $P_{Z|S=s} = P_Z$ *for* P_S*-almost every* s.

(iv) For every bounded measurable $f : \mathcal{Z} \to \mathbb{R}$, $\mathbb{E}[f(Z) \mid S = s] = \mathbb{E}[f(Z)]$ *for* P_S*-almost every* s.

Proof. See Appendix SM1.5 of the supplement. ∎

Remark 3.14. Theorem 3.13 confirms that zero HSIC corresponds to per-subgroup demographic parity simultaneously for every prediction head—the strongest population-level guarantee one can ask of an aggregate measure. In the non-zero regime, however, neither HSIC nor any integral measure $\mathcal{I}_d$ delivers worst-case subgroup guarantees; those require a sup-style measure $\sup_s d(P_{Z|S=s}, P_Z) \le \epsilon$, which is orthogonal to the joint-distribution argument here.

Two technical augmentations — a necessity-for-counterfactual-fairness statement (Proposition SM6.1) and a uniform-concentration bound that yields a train-test fairness bridge (Proposition SM6.2, Corollary SM6.3, Remark SM6.4) — are deferred to Sections SM6.1 and SM6.2 of the supplement.

3.5. The FRHSIC objective. A key advantage of HSIC over EIPM is the simplicity of its empirical estimation. Given i.i.d. samples $\{(Z_i, S_i)\}_{i=1}^n$, the biased empirical estimator of HSIC is (Gretton et al., 2005):

$$\widehat{\mathrm{HSIC}}(Z, S) = \frac{1}{n^2}\mathrm{tr}(KHLH), \tag{3.7}$$

where $K_{ij} = k_\mathcal{Z}(Z_i, Z_j)$, $L_{ij} = k_\mathcal{S}(S_i, S_j)$, and $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ is the centering matrix.

This estimator has the following advantages over the EIPM estimator of Kong et al. (2025):

- **No conditional smoothing:** EIPM estimation requires kernel-smoothed weights $\hat{w}_\gamma(j; i)$ to approximate $P_{Z|S=S_i}$, which introduces a bandwidth parameter γ and requires careful tuning. HSIC requires no such step.
- **Closed-form computation:** The HSIC estimator is a simple matrix trace, computable in $O(n^2)$ time. EIPM estimation involves computing a weighted

MMD for each sample point, leading to $O(n^3)$ complexity in the naïve implementation.

- **No leave-one-out correction:** The EIPM estimator of Kong et al. (2025) uses a leave-one-out construction $\hat{\mathbb{P}}^{(-i),\gamma}_{Z|S=S_i}$ for technical tractability. The HSIC estimator requires no such construction.
- **Well-studied convergence properties:** The convergence rate of the HSIC estimator is $O_p(1/\sqrt{n})$ (Proposition 3.15), which is independent of the dimensionality of S.

We state the convergence guarantee formally.

PROPOSITION 3.15 (Convergence of HSIC estimator (Gretton et al., 2005)). *Suppose the kernels $k_{\mathcal{Z}}$ and $k_{\mathcal{S}}$ are bounded: $k_{\mathcal{Z}}(z,z) \le \kappa_{\mathcal{Z}}$ and $k_{\mathcal{S}}(s,s) \le \kappa_{\mathcal{S}}$ for all $z \in \mathcal{Z}$, $s \in \mathcal{S}$. Then for any $\delta > 0$, with probability at least $1-\delta$:*

$$\left|\widehat{\mathrm{HSIC}}(Z,S) - \mathrm{HSIC}(Z,S)\right| \le O\left(\kappa_{\mathcal{Z}}\kappa_{\mathcal{S}}/\sqrt{n}\right) + O\left(\sqrt{\log(1/\delta)/n}\right).$$

In particular, $\widehat{\mathrm{HSIC}}(Z,S) \to \mathrm{HSIC}(Z,S)$ at rate $O_p(n^{-1/2})$, independent of $\dim(\mathcal{Z})$ and $\dim(\mathcal{S})$.

Remark 3.16. Kong et al. (2025) show that the EIPM estimator converges at rate $O_p(\gamma_n^2 + n^{-2/5})$ under the optimal kernel-smoothing bandwidth $\gamma_n \propto n^{-1/5}$. The HSIC convergence rate $O_p(n^{-1/2})$ is faster and involves no bandwidth parameter, reflecting the advantage of avoiding conditional smoothing. We empirically validate this convergence rate in Figure 1.

The following corollary makes the statistical efficiency advantage of HSIC explicit.

COROLLARY 3.17 (Statistical efficiency of HSIC vs. EIPM). *Under the assumptions of Proposition 3.15 and Theorem 4 of Kong et al. (2025), the HSIC estimator converges to its population value at rate $O_p(n^{-1/2})$, while the EIPM estimator converges at rate $O_p(n^{-2/5})$. Hence the HSIC estimator achieves the parametric rate, whereas the EIPM estimator does not.*

Proof. See Appendix SM1.6 of the supplement. □

Remark 3.18. The biased HSIC estimator is a bounded V-statistic and therefore has the standard $O_p(n^{-1/2})$ convergence rate in the nondegenerate case (Gretton et al., 2005). Our statistical comparison with EIPM should be interpreted at the estimator level: HSIC estimates a joint-dependence functional directly, while EIPM estimators require smoothing over the continuous sensitive attribute; the rates are for two different population functionals that are zero-equivalent at independence.

Remark 3.19 (Biased vs. unbiased HSIC estimator). Song et al. (2012) proposed an unbiased HSIC estimator based on U-statistics. We use the biased estimator (3.7) for two reasons. First, the biased estimator is always non-negative, ensuring the regularization term $\lambda \cdot \widehat{\mathrm{HSIC}}$ is a valid penalty. The unbiased estimator can be negative for finite samples, which creates optimization difficulties when used as a regularizer (the optimizer may exploit negative values rather than driving dependence to zero). Second, the biased estimator has lower variance for small batch sizes (Gretton et al., 2012), which is relevant since we compute HSIC on mini-batches of size $m = 256$. Both estimators are consistent and converge at rate $O_p(n^{-1/2})$.

Training objective.. A fair and sufficient representation can be obtained by solving

$$\min_{h,f} \mathcal{L}_{\mathrm{sup}}(f \circ h) + \lambda \cdot \widehat{\mathrm{HSIC}}(h(X), S), \tag{3.8}$$

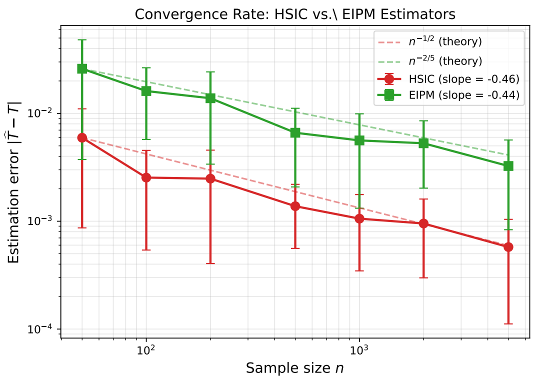

FIG. 1. *Empirical convergence of the HSIC and EIPM estimators as a function of sample size n (log–log scale). The HSIC estimator achieves the predicted $O(n^{-1/2})$ rate (slope ≈ -0.5), while the EIPM estimator converges at $O(n^{-2/5})$.*

where $\mathcal{L}_{\text{sup}}$ is the supervised loss (e.g., cross-entropy for classification, MSE for regression) and $\lambda > 0$ controls the fairness–accuracy tradeoff.

This can be viewed as a Lagrangian relaxation of the constrained problem

$$\min_{h,f} \ \mathcal{L}_{\text{sup}}(f \circ h) \quad \text{s.t.} \quad \text{HSIC}(h(X), S) \le \delta.$$

We parameterize h and f as neural networks and optimize Equation (3.8) using mini-batch stochastic gradient descent. In each mini-batch of size m, the HSIC term is estimated via Equation (3.7) using only the samples in the batch, yielding an $O(m^2)$ per-iteration cost. For kernel bandwidths, we use the median heuristic (Gretton et al., 2012): $\sigma_{\mathcal{S}}$ is computed once from the training data, and $\sigma_{\mathcal{Z}}$ is recomputed periodically from current mini-batch representations to track the evolving representation geometry.

Remark 3.20 (Mini-batch HSIC as a biased gradient surrogate). At fixed batch size $m < n$, the per-batch quantity $\widehat{\text{HSIC}}(Z_b, S_b) = m^{-2}\,\text{tr}(K_b H_b L_b H_b)$ is a V-statistic on a sub-sample, not an unbiased estimator of $\text{HSIC}(Z, S)$ on the full data; in particular, its gradient with respect to encoder parameters is a biased estimator of the population gradient at fixed m. We use this batch-level surrogate as a computational tool: it provides a tractable training signal that vanishes precisely when $\widehat{\text{HSIC}} \to 0$ on the full data, but the optimization dynamics are not those of standard unbiased SGD on the population HSIC. Empirically (Section 4), this batch-level signal is sufficient to drive the empirical $\widehat{\text{HSIC}}(Z, S)$ on the full data toward zero across all λ values we examine; we do not establish formal convergence guarantees.

3.6. Equal Opportunity and multiple-sensitive-attribute extensions. The HSIC-based approach naturally extends to Equal Opportunity (Madras et al., 2018), which requires fairness conditional on the positive outcome ($Y = 1$).

One simply computes the HSIC term using only the samples with $Y = 1$:

$$\widehat{\text{HSIC}}_{\text{EO}}(Z, S) = \frac{1}{n_1^2}\text{tr}(K_1 H_1 L_1 H_1),$$

where K_1, L_1, H_1 are the kernel and centering matrices restricted to the n_1 samples with $Y_i = 1$.

3.7. Validation-based regularization selection. A practical challenge in FRL is selecting the regularization strength λ. We use the HSIC independence test (Gretton et al., 2005) as a validation-based stopping criterion.

The HSIC test statistic under the null hypothesis $H_0 : Z \perp S$ follows a known distribution (a weighted sum of chi-squared variables). We use the gamma approximation (Gretton et al., 2005): under H_0, $n \cdot \widehat{\mathrm{HSIC}}$ is approximately gamma-distributed with parameters estimated from data.

Algorithm: HSIC-test-based λ selection.

1. **Input:** sample $\{(X_i, S_i, Y_i)\}_{i=1}^n$; kernels $k_{\mathcal{Z}}, k_{\mathcal{S}}$; increasing λ grid $\Lambda = \{\lambda^{(1)} < \lambda^{(2)} < \cdots < \lambda^{(K)}\}$; significance level α_{test} (e.g., 0.05).
2. **For** each $\lambda^{(j)} \in \Lambda$ in increasing order:
   (a) Train FRHSIC at $\lambda^{(j)}$ to obtain encoder $\hat{h}_{\lambda^{(j)}}$.
   (b) Compute $\widehat{\mathrm{HSIC}}(\hat{h}_{\lambda^{(j)}}(X), S)$ on a held-out validation split.
   (c) Run the gamma-approximated HSIC independence test at level α_{test} (Gretton et al., 2005).
   (d) **If** the test fails to reject $H_0 : Z \perp S$, **return** $\lambda^{(j)}$.
3. **Output:** the smallest $\lambda^{(j)}$ at which the test fails to reject independence (or $\lambda^{(K)}$ if no such λ is found).

This selects $\lambda^{(j)}$ such that the learned representation is statistically indistinguishable from independence at the chosen significance level. We view this as a practical validation heuristic based on a familiar independence test, not as a principled selection rule. On large validation sets the test can be overpowered, in which case we recommend combining it with an effect-size threshold on $\widehat{\mathrm{HSIC}}$ or applying the test on a fixed-size validation subsample (the practical-significance variant below).

Practical significance variant. On large datasets, the independence test may be overpowered: it detects statistically significant but practically negligible dependence. In such settings, we recommend a *practical significance* variant that combines the statistical test with a threshold on the HSIC value itself:

$$\text{stop at the smallest } \lambda^{(j)} \text{ such that } \widehat{\mathrm{HSIC}} < \epsilon \text{ or the test fails to reject at level } \alpha_{\text{test}},$$

where $\epsilon > 0$ is a user-specified practical significance threshold (e.g., $\epsilon = 10^{-3}$). This ensures the procedure remains useful regardless of sample size. We validate both variants on all five datasets in Section SM7.5.

4. Experiments. We evaluate FRHSIC on both synthetic and real-world data. All experiments use PyTorch with the Adam optimizer (learning rate 10^{-3}). Kernel bandwidths are set via the median heuristic (Gretton et al., 2012). Code is available at https://github.com/Yijin911/Replacement-for-EIPM (the repository will be made publicly accessible at submission; a fully archived release with a DOI will accompany the camera-ready version).

4.1. Synthetic Data: Validating the Theory.

Setup. We generate $n = 5000$ samples with $S \sim \mathrm{Uniform}(0, 1)$, $X = (S + \epsilon_1, \epsilon_2)$ where $\epsilon_1 \sim \mathcal{N}(0, 0.09)$, $\epsilon_2 \sim \mathcal{N}(0, 1)$, and $Y \sim \mathrm{Bernoulli}(1/(1+e^{-X_1}))$. The encoder h is a 2-layer MLP (hidden dimension 64, representation dimension 8), and the prediction head f is linear. (We use a smaller architecture than the real-data experiments to isolate the effect of the HSIC penalty in a controlled setting.) We train with objective $\mathcal{L}_{\mathrm{CE}}(f \circ h) + \lambda \cdot \widehat{\mathrm{HSIC}}(Z, S)$ for $\lambda \in \{0, 0.1, 0.5, 1, 5, 10, 50\}$.

Results. Figure 2 shows the results. As λ increases, GDP decreases monotonically toward zero (panel a), confirming that controlling HSIC effectively enforces

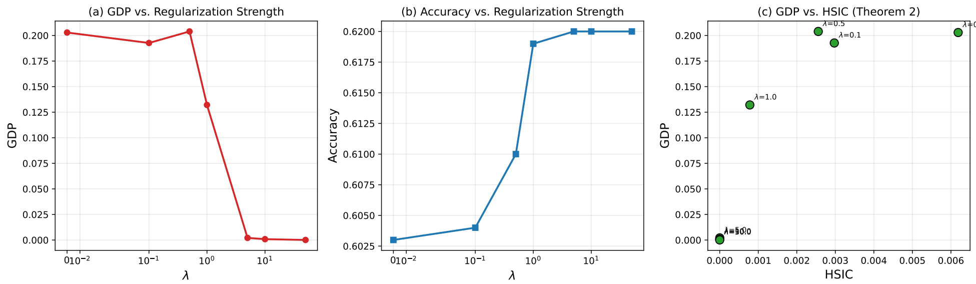

FIG. 2. *Synthetic experiment results. (a) GDP decreases as λ increases. (b) Accuracy remains stable across regularization strengths. (c) GDP vs. HSIC shows a monotone relationship, consistent with the conditional-gap-control interpretation of Theorem 3.4 (the plot displays monotonic dependence, not the LHS/RHS of the bound). The in-image panel label refers to the same result under the prior numbering ("Theorem 2"); the corresponding statement in the current numbering is Theorem 3.4. The overlapping λ tick labels in panel (a) will be regenerated for camera-ready.*

demographic parity. Panel (c) plots GDP against HSIC directly, showing that both decrease together in a monotone relationship consistent with the upper bound in Theorem 3.4. Notably, accuracy remains stable (around 0.62) even as fairness is strongly enforced, suggesting that the sensitive attribute contributes primarily to unfair discrimination rather than useful predictive signal in this setting.

4.2. Real Data Analysis.

4.2.1. Datasets. We evaluate on five real-world datasets with continuous sensitive attributes, spanning income, criminal justice, healthcare, and community-level outcomes:

- **Adult** (classification): $n \approx 30{,}000$ individuals; target is income >50K; sensitive attribute is age.
- **ACS Income** (classification): $n = 20{,}000$ individuals from the 2018 American Community Survey (California) via folktables (Ding et al., 2021); target is income >50K; sensitive attribute is age. This is the modern replacement for the UCI Adult dataset, with richer features and standardized preprocessing.
- **MEPS** (classification): $n \approx 13{,}000$ respondents from the 2015 Medical Expenditure Panel Survey (Panel 19); target is healthcare utilization ≥ 10 visits; sensitive attribute is age. This dataset is standard in the algorithmic fairness literature (Romano et al., 2020).
- **Crime** (regression): $n \approx 2{,}000$ communities; target is violent crimes per population; sensitive attribute is racial composition.
- **COMPAS** (classification): $n \approx 6{,}000$ defendants; target is two-year recidivism; sensitive attribute is age.

All features are min-max scaled to $[0, 1]$. The sensitive attribute S is also scaled to $[0, 1]$. Full reproducibility details, hyperparameter grids, per-dataset preprocessing, hardware, and the FREM 32-anchor subsampling approximation are documented in Section SM4 of the Supplementary Material.

4.2.2. Comparison with Continuous Sensitive Attribute Methods. Following Kong et al. (2025), we compare against both (1) regularization methods for continuous sensitive attributes and (2) FRL methods for binary attributes applied via binning:

- **Reg-GDP** (Jiang et al., 2022): regularizes GDP directly via kernel-smoothed conditional expectations.

- **FREM** (Kong et al., 2025): uses the weighted EIPM estimator with MMD as the discriminator.
- **ADV** (Grari et al., 2022): adversarial approach that trains an adversary to predict S from Z.
- **LAFTR (binned)** (Madras et al., 2018): adversarial FRL applied to binned (quartile) S.
- **MMD (binned)** (Deka and Sutherland, 2023): MMD-based FRL applied to binned S.
- **dCor** (Székely et al., 2007; Huo and Székely, 2016): regularizes distance covariance between Z and S, a kernel-free dependence measure with the same $O(n^2)$ cost as HSIC.
- **Unfair**: no fairness constraint ($\lambda = 0$).

To ensure a fair comparison, all methods use the same architecture as Kong et al. (2025): a 2-layer encoder with SELU activation (hidden dimension 50, representation dimension 50) and a linear prediction head. Each method is evaluated under the bandwidth convention from its source publication: FRHSIC uses the median heuristic of Gretton et al. (2012), the standard choice for HSIC; FREM uses a fixed $\sigma_Z = 1.0$ for its Z-kernel and a smoothing bandwidth $\gamma = 0.5$ on S (Kong et al., 2025); Reg-GDP uses a fixed $\sigma_Z = 1.0$ for its Z-kernel and a Nadaraya–Watson smoothing bandwidth 0.2 on S for the conditional expectation (Jiang et al., 2022). For computational tractability we subsample 32 anchor points per batch for the FREM EIPM computation; this approximation is discussed further in Appendix SM4. We do not retune bandwidths to favor any method. Models are trained for 200 epochs with Adam (learning rate 10^{-3}). We sweep $\lambda \in \{0.1, 1.0, 10.0, 100.0, 500.0\}$ and report results over 5 random 80%/20% train/test splits (mean $\pm$ standard deviation). Table 2 reports the $\lambda = 10$ slice; Pareto frontiers across all λ are shown in Figure 3.

Results are presented in Table 2 and Figures 3–4.

Fairness–prediction tradeoff (Table 2, Figure 3).. FRHSIC and FREM occupy different points on the fairness–accuracy Pareto frontier. FRHSIC tends to preserve accuracy more aggressively at moderate λ, while FREM and Reg-GDP can drive GDP lower at greater accuracy cost; the choice depends on which corner of the frontier the application requires. On Adult, FRHSIC with adaptive bandwidth achieves GDP $= 0.511$ at $\lambda = 10$ while preserving accuracy (0.852, matching Unfair at 0.849); FREM achieves lower GDP (0.461) at slightly lower accuracy (0.847), and the FRHSIC training is 36$\times$ faster per epoch. Reg-GDP drives GDP to ≈ 0 but accuracy drops to 0.758. On Crime, FRHSIC attains MSE $= 1.96$ (matching Unfair at 1.99) at GDP $= 0.030$, while FREM reaches GDP $= 0.015$ at MSE $= 2.14$ and Reg-GDP reaches GDP ≈ 0.002 at MSE $= 4.61$. On COMPAS, FRHSIC preserves the strongest accuracy among the methods listed (0.661) while reducing GDP from the unfair baseline; FREM and Reg-GDP achieve lower GDP at greater accuracy cost (Reg-GDP collapses to 0.545, near random). Notably, FRHSIC matches or slightly exceeds the Unfair baseline in prediction performance across all five datasets. This suggests that, on these datasets, the HSIC penalty may also act as a regularizer. We do not rely on this effect for the main fairness claim. The dCor baseline behaves similarly to FRHSIC (both are joint independence measures), with FRHSIC achieving lower MI. Both HSIC (with characteristic kernels) and distance covariance characterize independence, so the choice between them is largely empirical: FRHSIC is broadly competitive with dCor as a regularizer, and HSIC has the direct RKHS–MMD/EIPM connection used in our theory, while the median heuristic provides a default bandwidth choice, reducing one tuning burden.

Representation fairness (Table 2, Figure 4).. Reg-GDP regularizes a head-specific GDP loss (the prediction g on the training task), so vanishing Reg-GDP makes that head fair without controlling other downstream heads. EIPM/FREM regularizes a representation-level discrepancy on Z, controlling all RKHS heads simultaneously; FRHSIC is also representation-level, since HSIC penalizes all dependence between Z and S and any downstream predictor built on Z inherits the fairness property. This distinction matters for interpreting GDP scores. Reg-GDP achieves GDP = 0 on Adult, but this only means the *specific linear head used during training* is fair—the representation itself still leaks S. Indeed, Reg-GDP retains $\mathrm{MI}(Z, S) = 0.017$, meaning a downstream adversary could train a new head on the same representation and discriminate based on S. By contrast, FRHSIC at $\lambda = 10$ achieves $\mathrm{MI}(Z, S) = 0.008$—a 10× reduction from the Unfair baseline (0.080)—while preserving accuracy (0.852 vs. Reg-GDP's 0.758). FRHSIC's higher GDP (0.511 vs. 0.000) at the same λ reflects that it distributes its regularization budget across *all forms of dependence* rather than concentrating it on first-moment matching for a single head. At higher regularization ($\lambda = 500$), FRHSIC drives GDP to 0.457 while maintaining accuracy at 0.850 (Figure 3), approaching Reg-GDP's GDP at dramatically better accuracy. The key insight is that comparing methods at the same λ is misleading, because HSIC and GDP penalties have different scales; the Pareto frontier (Figure 3) provides a fairer comparison, where FRHSIC achieves a competitive fairness–accuracy tradeoff while additionally guaranteeing representation-level fairness. On Crime, FRHSIC achieves MI = 0.015 (an 11.8× reduction from the Unfair baseline at 0.177) with *no MSE degradation*. Reg-GDP achieves lower MI only by sacrificing 2.3× worse MSE.

A note on MI estimates for adversarial methods. MI is reported as a diagnostic and not a primary fairness audit. The KSG estimator we use can underestimate MI for high-dimensional learned representations whose dependence structure violates the smoothness assumptions underlying nearest-neighbor entropy estimation (Gretton et al., 2012), and we therefore interpret MI together with HSIC, GDP, and downstream prediction performance rather than in isolation. ADV and LAFTR illustrate this: they report MI = 0.000 on Adult, yet their GDP values remain high (0.590 and 0.462, respectively). Since MI = 0 would imply $Z \perp S$ and hence GDP = 0 for any head, these MI values are inconsistent with the observed GDP, suggesting that adversarial training produces representations whose dependence structure (e.g., concentrated on low-dimensional manifolds) is poorly suited to nearest-neighbor entropy estimation. By contrast, FRHSIC's MI estimate (0.008) is consistent with its GDP (0.511): the representation retains some dependence on S, but substantially less than the Unfair baseline. We treat this consistency as supporting evidence rather than a fairness certificate.

Hyperparameter burden. The conditional-estimation route comes with additional knobs that FRHSIC does not need: FREM has both a smoothing bandwidth γ_n on S and an anchor count for its weighted-MMD approximation (we use the 32-anchor subsampling default; the sensitivity of FREM to these choices was not exhaustively swept here). Reg-GDP has its own Nadaraya–Watson bandwidth on S. FRHSIC uses the median heuristic for both kernels and has no smoothing bandwidth on S; the only fairness-specific hyperparameter is the regularization strength λ, which is selected via the validation-based procedure of Section 3.7.

Computational advantage. FRHSIC is ∼36× faster than FREM per training epoch (Section 4.3), requires no bandwidth tuning for conditional smoothing, and uses a simple $O(m^2)$ per-batch estimator. We note that Reg-GDP is faster still (1.69s

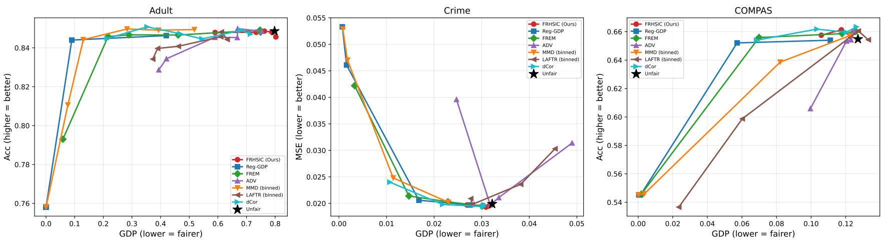

FIG. 3. *Fairness–accuracy Pareto frontiers on real-world datasets. Each point corresponds to a different λ value. FRHSIC (red) attains a competitive fairness–accuracy tradeoff relative to the baselines considered.*

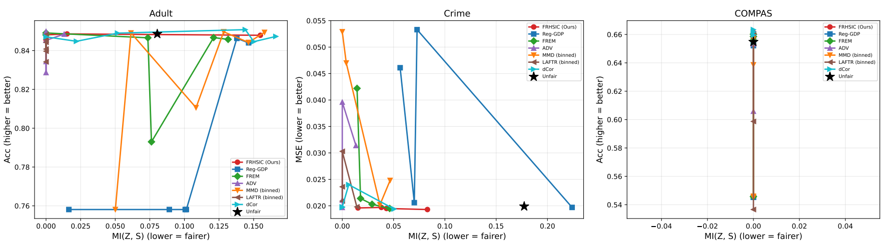

FIG. 4. *Representation fairness:* $\mathrm{MI}(Z, S)$ *vs. prediction performance. FRHSIC tends to attain lower* $\mathrm{MI}(Z, S)$ *than regularization methods (Reg-GDP) at comparable accuracy in these experiments, consistent with the learned representation carrying less information about* S*.*

vs. 2.82s per epoch at $n = 20{,}000$), since it evaluates conditional expectations on a fixed grid rather than computing kernel matrices. The computational advantage of FRHSIC is thus relative to EIPM-based methods (FREM), which are the most directly comparable approach (both enforce distributional fairness criteria via kernel embeddings).

4.2.3. Comparison with FRL Methods for Binary Attributes. To compare with binary-attribute FRL methods, we bin the continuous sensitive attribute into quartiles and apply LAFTR (Madras et al., 2018). As shown in Table 2, LAFTR consistently achieves worse fairness–accuracy tradeoffs than the continuous methods (FRHSIC, FREM, Reg-GDP), confirming the advantage of treating S as continuous rather than discretizing it.

Pareto summary. For a single-number summary of the fairness–accuracy tradeoff, we report for each method the lowest GDP it attains while preserving accuracy within 1% of the unfair baseline (Table SM1 in the supplement, computed from the Pareto sweep). On Adult, FRHSIC and FREM achieve comparable values, with FREM slightly lower; on Crime, FRHSIC achieves the lowest GDP within the accuracy band; on COMPAS, Reg-GDP achieves a lower number but at substantial accuracy cost (it falls outside the 1% band). The single-number summary should be read alongside the full Pareto frontiers (Figure 3); no method dominates across all datasets.

Transfer to held-out heads. A representation-level fairness penalty (HSIC, EIPM) should yield representations on which *any* downstream head is fair. We verify this by training a single representation per method (at fixed λ) and then training fresh prediction heads of multiple architectures (linear, 2-layer MLP, random forest, SVM) on the frozen representation. Table 3 reports the standard deviation of GDP across the four head architectures (lower = more head-invariant fairness); the full per-head

TABLE 2

Test performance and fairness metrics at $\lambda = 10$. Acc for classification datasets (higher = better); MSE $\times 10^2$ for Crime (lower = better). GDP: lower = fairer. MI: $\mathrm{MI}(Z, S)$ estimated via KSG, lower = fairer representation (— where the KSG estimator is unreliable due to the low-dimensional, discrete-valued feature space). Bold entries indicate the best value per (dataset, metric) cell among fairness methods (excluding the Unfair baseline). Bandwidth conventions follow each method's standard practice: FRHSIC and dCor use the median heuristic of Gretton et al. (2012) for the kernels they apply to Z; FREM uses a fixed $\sigma_Z = 1.0$ for its Z-kernel and a smoothing bandwidth $\gamma = 0.5$ on S (Kong et al., 2025); Reg-GDP uses a fixed $\sigma_Z = 1.0$ for its Z-kernel and a Nadaraya–Watson smoothing bandwidth 0.2 on S for the conditional expectation (Jiang et al., 2022). All bandwidths are values reported by their source publications.

Dataset	Method	Perf.	GDP (↓)	MI (↓)
Adult (Acc ↑)	Unfair	$.849 \pm .002$	$.798 \pm .068$	.080
	FRHSIC (Ours)	$\mathbf{.852 \pm .004}$	$.511 \pm .021$	**.008**
	FREM	$.847 \pm .002$	$.461 \pm .028$	.074
	Reg-GDP	$.758 \pm .003$	$\mathbf{.000 \pm .000}$	.017
	ADV	$.845 \pm .003$	$.590 \pm .058$	.000
	MMD (binned)	$.850 \pm .003$	$.283 \pm .027$	.128
	LAFTR (binned)	$.841 \pm .009$	$.462 \pm .133$	.000
	dCor	$.845 \pm .008$	$.545 \pm .018$	.150
ACS Income (Acc ↑)	Unfair	$.789 \pm .006$	$.482 \pm .032$	.158
	FRHSIC (Ours)	$\mathbf{.789 \pm .006}$	$.412 \pm .036$	**.078**
	FREM	$.786 \pm .004$	$.253 \pm .041$	.066
	Reg-GDP	$.590 \pm .008$	$\mathbf{.000 \pm .000}$	.094
	ADV	$.782 \pm .003$	$.265 \pm .015$	.000
	MMD (binned)	$.785 \pm .005$	$.128 \pm .023$	.055
	LAFTR (binned)	$.740 \pm .032$	$.137 \pm .019$	.000
	dCor	$.791 \pm .004$	$.394 \pm .049$	.137
MEPS (Acc ↑)	Unfair	$.802 \pm .006$	$.608 \pm .030$	.215
	FRHSIC (Ours)	$\mathbf{.803 \pm .007}$	$.508 \pm .020$	**.147**
	FREM	$.790 \pm .003$	$.089 \pm .016$	.000
	Reg-GDP	$.784 \pm .005$	$\mathbf{.000 \pm .000}$	.208
	ADV	$.790 \pm .004$	$.119 \pm .009$	.000
	MMD (binned)	$.785 \pm .005$	$.057 \pm .022$	.000
	LAFTR (binned)	$.789 \pm .002$	$.113 \pm .063$	.000
	dCor	$.804 \pm .004$	$.481 \pm .037$	.121
Crime (MSE ↓)	Unfair	$1.99 \pm .05$	$.032 \pm .004$	.177
	FRHSIC (Ours)	$\mathbf{1.96 \pm .04}$	$.030 \pm .004$	**.015**
	FREM	$2.14 \pm .07$	$.015 \pm .005$	.018
	Reg-GDP	$4.61 \pm .17$	$\mathbf{.002 \pm .000}$	.056
	ADV	$3.14 \pm .38$	$.049 \pm .019$	.013
	MMD (binned)	$4.70 \pm .19$	$.002 \pm .000$	.004
	LAFTR (binned)	$2.36 \pm .21$	$.038 \pm .015$	.000
	dCor	$1.98 \pm .06$	$.022 \pm .006$	.000
COMPAS (Acc ↑)	Unfair	$.655 \pm .017$	$.127 \pm .019$	—
	FRHSIC (Ours)	$\mathbf{.661 \pm .013}$	$.118 \pm .015$	—
	FREM	$.656 \pm .010$	$.070 \pm .025$	—
	Reg-GDP	$.545 \pm .009$	$\mathbf{.001 \pm .001}$	—
	ADV	$.653 \pm .003$	$.121 \pm .025$	—
	MMD (binned)	$.546 \pm .008$	$.004 \pm .002$	—
	LAFTR (binned)	$.599 \pm .011$	$.060 \pm .015$	—
	dCor	$.662 \pm .019$	$.103 \pm .044$	—

breakdown is in Section SM7.1 of the supplement. FRHSIC's cross-head GDP variance is comparable to FREM and substantially lower than the Unfair baseline; Reg-GDP attains a lower variance number but at substantial accuracy cost on Adult (accuracy collapses to 0.758; see Table 2). Transferability across downstream heads, not just a single trained head, is one of the principal reasons fairness should be enforced at the representation level.

TABLE 3

Cross-head GDP standard deviation (lower = more head-invariant fairness): standard deviation of GDP across four prediction heads (linear, MLP, random forest, SVM) trained on a single frozen representation at $\lambda = 10$ (Adult) and $\lambda = 100$ (Crime). See Section SM7.1 of the supplement for the full per-head table and figure.

Method	Adult ($\lambda = 10$)	Crime ($\lambda = 100$)
Unfair	0.348	—
FRHSIC (Ours)	0.244	0.001
FREM	0.247	0.002
Reg-GDP	0.058 †	0.002
ADV	0.252	0.005

† Reg-GDP attains lower cross-head variance on Adult but with accuracy collapsing to 0.758 (Table 2).

Significance. Mean and standard deviation across the 5 random splits are reported in Table 2. We do not report formal paired significance tests because $n = 5$ paired observations is too few for reliable inference at conventional significance levels; differences within ± 0.01 on accuracy or ± 0.05 on GDP should be read as practically equivalent.

COMPAS age as ordinal-with-repeats. COMPAS age is recorded as integer-valued and takes only a small number of distinct values, making it ordinal-with-repeats rather than truly continuous; this matters for the rank of $\widetilde{L}$ and hence for the Theorem 3.4 bound (Section 3).

4.3. Computational Efficiency. We compare the per-epoch training time of FRHSIC, FREM, and Reg-GDP on synthetic data with varying sample sizes $n \in \{500, 1000, 2000, 5000, 10000, 20000\}$. All methods use the same encoder, prediction head, and mini-batch size $m = 256$. Figure 5 shows the results. Per batch, FRHSIC computes a single $m \times m$ kernel matrix and a matrix trace, costing $O(m^2)$. FREM computes a weighted MMD for each of the m samples in the batch, requiring $O(m)$ vector-matrix products per sample, yielding $O(m^3)$ per batch. Reg-GDP evaluates conditional expectations on a fixed grid, making it the fastest. As n grows (more batches per epoch), FRHSIC retains a constant per-batch advantage over FREM, resulting in faster training across the sample sizes we tested. At $n = 20{,}000$, FRHSIC completes an epoch in 2.82 seconds compared to 100.7 seconds for FREM—a $\sim 36\times$ speedup—while Reg-GDP takes 1.69 seconds. The $36\times$ headline reports measured per-epoch training time on Adult-scale data ($n \approx 30{,}000$, batch size 256). At full-batch scale this gap is consistent with the $O(n^2)$ vs. $O(n^3)$ asymptotics of the empirical HSIC and EIPM estimators; at mini-batch scale the per-batch cost is $O(m^2)$ for FRHSIC and is dominated for FREM by the weighted-MMD computation, which itself includes a 32-anchor subsampling for tractability. We do not claim that mini-batch FRHSIC literally scales as $O(n^2)$ versus $O(n^3)$ in wall-clock time — the asymptotic ratio describes full-batch estimator complexity, while the wall-clock ratio reflects implementation constants and the anchor-subsampling approximation.

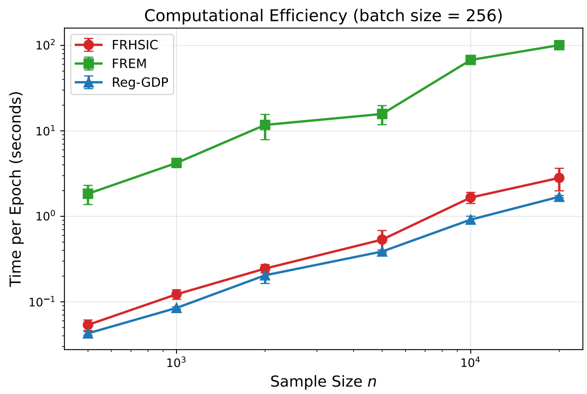

FIG. 5. *Per-epoch wall-clock training time as a function of dataset size, comparing FRHSIC and FREM. Both methods use the same neural-network architecture and mini-batch size; the gap reflects (i) the closed-form* $\mathrm{tr}(KHLH)$ *kernel-matrix update for FRHSIC versus FREM's per-batch weighted-MMD computation with* 32*-anchor subsampling, and (ii) the* $O(n^2)$ *vs.* $O(n^3)$ *full-batch estimator complexity that dominates at large* n*. Mini-batch wall-clock time at fixed batch size* m *scales as* $O(m^2)$ *for both methods; the* 36$\times$ *headline ratio is the measured per-epoch ratio at full-Adult scale (*$n \approx 3 \times 10^4$*, batch size* 256*), not a literal asymptotic claim.*

Additional experiments — a transfer study across downstream heads, an empirical tightness sweep for the Theorem 3.4 bound, a high-dimensional-sensitive-attribute comparison, ablations over kernel choice and representation dimensionality, and validation of the HSIC-independence-test λ-selection rule of Section 3.7 — are reported in Section SM7.1–SM7.5 of the supplement.

5. Related Work.

Fair Representation Learning. Zemel et al. (2013) introduced the concept of learning fair representations by mapping data to a latent space that preserves information about Y while obfuscating S. Louizos et al. (2015) extended this idea using variational autoencoders, and Madras et al. (2018) proposed adversarial training (LAFTR) to enforce fairness. Creager et al. (2019) introduced disentangled representations for flexible fairness constraints. Oneto et al. (2020) used MMD and Sinkhorn divergences, and Deka and Sutherland (2023) proposed MMD-B-Fair with statistical testing. Quadrianto et al. (2019) learned fair representations in the data domain. These methods are designed for categorical sensitive attributes and require binning when applied to continuous S, which incurs information loss (as our experiments confirm).

Fairness with Continuous Sensitive Attributes. Four widely-used criteria target fairness with continuous S: GDP (Jiang et al., 2022), estimated via kernel-smoothed conditional expectations (Reg-GDP); EIPM (Kong et al., 2025), estimated via weighted empirical IPM with bandwidth γ_n (FREM); kernel HGR / NLD (Mary et al., 2019; Grari et al., 2022), estimated via regularized kernel canonical correlation; and mutual-information-based criteria (Cho et al., 2020), estimated via density or copula approximations. Grari et al. (2022) additionally use an adversarial network (ADV) to approximate the kernel HGR through a learned discriminator. GDP, EIPM/FREM, and MI-based criteria are naturally written through conditional or density-ratio quantities and therefore require smoothing, density or copula approximation, or weighted-empirical estimation. Kernel HGR/NLD is different: it is a global-dependence criterion estimated through regularized kernel CCA, so its main practical issue is operator regularization and finite-sample stability rather than conditional smoothing on S. FRHSIC differs from both routes by using a joint V-statistic for $P_{Z,S}$ versus $P_Z \otimes P_S$, whose closed-form estimator achieves the parametric rate $O_p(n^{-1/2})$; we establish equivalence theorems that show, for the conditional-route criteria, the choice between routes is largely an estimation question.

HSIC in Machine Learning. HSIC was introduced by Gretton et al. (2005) and has since been used extensively for independence testing (Gretton et al., 2007; Albert et al., 2022), feature selection (Song et al., 2007, 2012), dimensionality reduction (Ma et al., 2018), and as a training objective for deep networks (Ma et al., 2020). The connection between HSIC and distance covariance (Sejdinovic et al., 2013) further situates our work within the broader literature on dependence measures. Specifically, distance covariance is equivalent to HSIC with a semimetric kernel of the form $k(x, y) = \|x-a\|+\|y-a\|-\|x-y\|$, which is characteristic but has no bandwidth parameter. Both HSIC (with characteristic kernels) and distance covariance characterize independence, so the choice between them is largely empirical. In our experiments (Table 2) FRHSIC is broadly competitive with dCor as a regularizer, and HSIC has the direct RKHS–MMD/EIPM connection used in our theory; the median heuristic provides a default bandwidth choice, reducing one tuning burden. While HSIC as a regularizer is well-studied, prior work has not, to our knowledge, formally connected it to the specific fairness measures (GDP, EIPM) developed in the continuous-S fairness literature. Our contribution makes this connection precise: Theorem 3.1 relates HSIC to the squared expected MMD, and Theorem 3.4 provides a spectral upper bound on a *projected* empirical conditional-gap surrogate (the unprojected gap is recovered under the rank-$(n-1)$ condition of Corollary 3.5), with the population-level HSIC–GDP implication recorded separately in Proposition 3.6. We additionally propose a regularization-strength selection procedure based on the HSIC independence test, and provide a systematic comparison against existing continuous-S methods (FREM, Reg-GDP, ADV) on a common benchmark. Lee et al. (2022) use MMD for fair PCA but do not address continuous sensitive attributes or the HSIC–GDP connection.

Position relative to prior HSIC work.. HSIC and related dependence penalties have been used in independence testing, representation learning, and fairness-motivated regularization (Gretton et al., 2007; Song et al., 2012; Ma et al., 2020; Albert et al., 2022). Our use of HSIC is different in emphasis: we use it to instantiate a joint-distribution formulation of fair representation learning with continuous sensitive attributes. This formulation avoids the conditional-estimation step required by GDP/EIPM-style criteria while preserving the same zero-level population independence target.

Causal Fairness. Kusner et al. (2017) proposed counterfactual fairness, requiring that predictions be invariant to interventions on S. In the structural causal model $S \to X \to Z \to \widehat{Y}$ (with possible $S \to Y$), counterfactual fairness requires that the path $S \to X \to Z$ carries no information about S. Our HSIC criterion enforces $Z \perp S$ (observational independence), which is a *natural prerequisite* for counterfactual fairness: if Z is observationally dependent on S, then some downstream predictor can discriminate based on S. Observational independence is not sufficient for counterfactual fairness in general (e.g., unobserved confounders can make the two notions diverge), but is necessary under structural assumptions: under the SCM in Section 2.8 with a deterministic encoder, $\mathrm{HSIC}(Z, S) = 0$ is implied by counterfactual fairness for every prediction head in a sufficiently rich class (Proposition SM6.1). HSIC therefore provides a tractable observational shadow of counterfactual fairness.

GDP and EIPM, by contrast, only block the path from S to $\widehat{Y}$ through a specific prediction head f—the representation Z may still encode S, enabling other heads to recover it. This distinction explains why FRHSIC achieves lower $\mathrm{MI}(Z, S)$ than Reg-GDP: FRHSIC removes dependence at the representation level, while Reg-GDP only masks it for a specific predictor.

Two routes to the same population target. The conceptual thread connecting our equivalence theorems is that HSIC implements a joint-distribution route to representation-level independence, formulating fairness as a property of $P_{Z,S}$ versus $P_Z \otimes P_S$, while the integral framework implements a conditional-distribution route, averaging or maximizing a discrepancy that depends on the family $\{P_{Z|S=s}\}_{s\in\mathcal{S}}$.

The equivalence theorems—HSIC $\Leftrightarrow$ EIPM (Proposition 3.9 and Theorem 3.1) and HSIC $\Rightarrow$ GDP at the population level (Proposition 3.6), together with the fact that population HSIC and HGR are both zero precisely when $Z \perp S$ under characteristic kernels—confirm that this distinction is one of estimation, not expressivity: the same fair representations are identified by either route. The estimation-rate gap ($O_p(n^{-1/2})$ for HSIC vs. $O_p(n^{-2/5})$ or worse for the conditional-estimation route) is the empirical reflection of this estimator-level asymmetry, with the caveat that it is a comparison between estimators of different (but population-zero-equivalent) functionals.

6. Discussion. FRHSIC is based on a joint-distribution formulation of representation-level independence. Instead of estimating the conditional family $\{P_{Z|S=s}\}_s$, it penalizes a joint-vs-product discrepancy between $P_{Z,S}$ and $P_Z \otimes P_S$. Our theoretical results establish that HSIC is tightly connected to existing fairness notions: it decomposes into a kernel-weighted aggregation of conditional MMD deviations (Theorem 3.1), upper-bounds an empirical second-moment conditional-gap quantity that dominates the squared empirical absolute prediction gap for any RKHS prediction head (Theorem 3.4), and zero population HSIC implies zero population GDP for every prediction head (Proposition 3.6). Beyond the population-level theory, we provide a principled regularization selection procedure via the HSIC independence test (Section 3.7). The resulting algorithm is computationally simpler, avoids bandwidth selection for conditional smoothing, and maintains competitive fairness–accuracy tradeoffs.

Extensions validated. We validate two extensions of the basic framework. First, the Equal Opportunity extension (Section 3.6), which computes HSIC on the $Y = 1$ subset, provides a principled approach to conditional fairness (Appendix SM5). Second, FRHSIC naturally extends to multiple continuous sensitive attributes by using a product kernel on the joint (S_1, S_2) space; preliminary experiments with age and hours-per-week on the Adult dataset confirm effectiveness (Appendix SM5).

Limitations and scope. FRHSIC does not make GDP or EIPM obsolete. These criteria remain natural when the desired fairness object is explicitly conditional or head-specific. HSIC and EIPM are zero-equivalent under characteristic kernels, but away from zero they are different functionals and need not rank representations identically. Our claim is that, when the goal is transferable representation-level independence with continuous sensitive attributes, the joint route avoids the conditional-estimation burden while preserving the same zero-level fairness target. Several technical limitations remain. First, the empirical bound in Theorem 3.4 scales as $\|f\|^2_{\mathcal{F}_\mathcal{Z}}/\hat{\lambda}_S$, and $\hat{\lambda}_S$ can be small in practice (cf. the empirical sweep in Section SM7.2), which makes the bound loose for narrow-bandwidth kernels. Adaptive kernel selection to enlarge $\hat{\lambda}_S$ while preserving the characteristic property, and tighter spectral arguments that exploit the full spectrum rather than only the smallest positive eigenvalue, are concrete paths to practically informative GDP guarantees. Second, HSIC enforces a stronger condition than GDP: it penalizes all forms of dependence between Z and S, not just the first-moment difference that GDP captures. While this ensures fairness across all prediction heads (as validated in Section SM7.1), it may be overly conservative in settings where only specific fairness notions are required. Third, the $O(n^2)$ cost of HSIC estimation, while lower than the $O(n^3)$ cost of EIPM, may still be prohibitive for very large datasets. Nyström approximations (Gretton et al., 2012) or random Fourier features could reduce this to $O(nm)$ for $m \ll n$. Fourth, the transferability guarantee ($Z \perp S$ implies fairness for any head) holds at the population level. In practice, we minimize *empirical* HSIC on training data, and the learned representation must generalize to unseen data; a uniform-concentration argument that bridges training and population HSIC, and hence delivers a train-test fairness guarantee, is given in Section SM6.2 of the supplement.

Connection to causal fairness. Proposition SM6.1 formalizes the relationship between HSIC-fairness and counterfactual fairness (Kusner et al., 2017): under the stated SCM, counterfactual fairness for every sufficiently rich prediction head implies $\mathrm{HSIC}(Z, S) = 0$. The converse does not hold, as observational and counterfactual independence diverge in the presence of structurally hidden dependence. HSIC therefore enforces the strongest observational shadow of counterfactual fairness — a tractable necessary condition that requires no SCM specification — while acknowledging that the full counterfactual criterion lies strictly beyond observational reach.

Auditing limitations. This paper reports HSIC, GDP, and a KSG estimate of MI as the main fairness audit metrics on the held-out test split. Additional held-out audits (e.g., distance correlation, adversarial-prediction-of-S baselines) would provide complementary evidence and are a natural extension; we have not run them here.

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